LIBBY — MANUSCRIPTS INVENTORY

# Manuscripts considered — tnbc-brca1-oligomet-liver-r7p3

*Master inventory: every paper reviewed in this case*

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# Manuscripts considered — tnbc-brca1-oligomet-liver-r7p3

Master inventory: 36 clinical (30 included, 6 considered & excluded) + 24 pre-clinical (22 included, 2 considered & excluded) + 10 additional trial publications/registrations. One entry per manuscript, sorted by year (newest first). Excluded entries are papers a Libby agent reviewed and chose NOT to feed to the board, with the exclusion reason captured.

### Tolaney SM/Rugo HS (2026) — N Engl J Med

**Reference:** — · **Type:** clinical · **Inclusion:** included

- **Intervention:** Sacituzumab govitecan + pembrolizumab
- **Indication / model:** Previously untreated PD-L1+ (CPS >=10) metastatic TNBC, 1L (1L); PD-L1 CPS >=10 by 22C3; treatment-naive in advanced setting; ECOG 0-1; both BRCA-mut and -WT eligible.
- **Design:** Randomized open-label phase 3 (ASCENT-04/KEYNOTE-D19)
- **n:** 443
- **Effect size:** 0.65 HR — PFS (BICR)
- **Variance:** 95% CI 0.51–0.84; p=<0.001
- **Toxicities:** neutropenia G>=3 (n=221, 50%); immune-related AE (n=221, 25%)
- **Case fit:** strong
- **Notes:** ASCENT-04 supports a 1L SG + pembrolizumab regimen for PD-L1 CPS >=10 mTNBC. Pivotal readout from 2025 presentations; full publication is the next decision-relevant pivot, currently in press at the time of writing — confirm citation when available.

### Dent R/Bardia A (2026) — Ann Oncol

**Reference:** — · **Type:** clinical · **Inclusion:** included

- **Intervention:** Datopotamab deruxtecan
- **Indication / model:** Previously untreated locally recurrent inoperable or metastatic TNBC ineligible for PD-1/PD-L1 inhibitors (e.g. PD-L1 CPS <10) (1L); 1L mTNBC, ICI-ineligible (PD-L1 CPS <10 by 22C3 or prior anti-PD-1 in early setting <6 mo prior); ECOG 0-1.
- **Design:** Randomized open-label phase 3 (TROPION-Breast02)
- **n:** 644
- **Effect size:** 0.57 HR — PFS (BICR)
- **Variance:** 95% CI 0.44–0.73; p=<0.0001
- **Toxicities:** stomatitis/mucositis G>=3 (n=323, 6%); ILD/pneumonitis (n=323, 3%); any treatment-related AE G>=3 (n=323, 35%)
- **Case fit:** partial
- **Notes:** TROPION-Breast02 is the 1L PD-L1-low / ICI-ineligible TNBC pivotal for Dato-DXd. If patient's PD-L1 CPS returns <10 she lands in this trial's target population. Anchors near-term label expansion.

### Rugo HS/Llombart-Cussac A (2026) — Clin Cancer Res

**Reference:**PMID 41405563 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Pembrolizumab + olaparib maintenance after pembro+chemo induction

- **Indication / model:** Locally recurrent inoperable or metastatic TNBC after induction with 1L pembrolizumab + chemo (maintenance); mTNBC with SD/CR/PR on 4-6 cycles of pembro+chemo induction; HRR-mut stratified.
- **Design:** Randomized open-label phase 2/3 (KEYLYNK-009)
- **n:** 271
- **Effect size:** 0.98 HR — PFS post-induction
- **Variance:** 95% CI 0.72–1.33; p=0.46
- **Toxicities:** any treatment-related AE G>=3 (n=135, 84.4%); any treatment-related AE G>=3 (n=136, 96.2%); anemia G>=3 (n=135, 25%)
- **Case fit:** partial
- **Notes:** Overall negative; BRCA1/2-mut subgroup signal (HR ~0.70) supports the strategy of pembro+chemo induction followed by olaparib-based maintenance for BRCA1-mut patients. Hypothesis-generating, not practice-changing.

### Dent R/Traina T (2026) — Ann Oncol

**Reference:**PMID 41937088 · **Type:** trial · **Inclusion:** included

- **Intervention:** Datopotamab deruxtecan 6 mg/kg IV q3w
- **Indication / model:** Previously untreated locally recurrent inoperable or metastatic triple-negative breast cancer ineligible for PD-1/PD-L1 inhibitors (1L); PD-L1 CPS <10 by 22C3 or other ICI ineligibility
- **Design:** randomized open-label phase 3 (TROPION-Breast02)
- **n:** 644
- **Effect size:** PFS HR 0.57; median PFS 10.8 vs 5.6 mo — PFS and OS dual primary
- **Variance:** 95% CI 0.46–0.71; p=<0.0001
- **Case fit:** partial
- **Notes:** (toxicity table not extracted)

### Wang J/— (2026)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** Inavolisib (GDC-0077) + eribulin
- **Indication / model:** PIK3CA-mutant metastatic triple-negative breast cancer (2L+); PIK3CA hotspot mutation (presumed therascreen-equivalent)
- **Design:** single-arm phase 2
- **n:** 40
- **Effect size:** — (not yet recruiting) — ORR / PFS
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

### Schmid P/Turner NC (2025) — Ann Oncol

**Reference:**doi:10.1016/j.annonc.2025.10.025 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Capivasertib + paclitaxel (CAPItello-290)
- **Indication / model:** Previously untreated metastatic TNBC (1L); 1L mTNBC; dual primary endpoint in overall and PIK3CA/AKT1/PTEN-altered (30.7%) populations; ECOG 0-1.
- **Design:** Randomized double-blind placebo-controlled phase 3 (CAPItello-290)

- **n:** 812
- **Effect size:** 0.92 HR — OS overall and AKT-pathway-altered populations (dual primary)
- **Variance:** p=0.32
- **Toxicities:** diarrhea G>=3 (n=400, 15%); hyperglycemia G>=3 (n=400, 3%); rash G>=3 (n=400, 5%)
- **Case fit:** partial
- **Notes:** Negative phase 3 OS readout in TNBC, including in the PIK3CA/AKT1/PTEN-altered stratum that would have been the patient's most likely benefit subgroup. Decision-relevant evidence against off-label capivasertib + paclitaxel in 1L mTNBC.

### Schmid P/Curigliano G (2025) — Ther Adv Med Oncol

**Reference:**PMID 40297626 · **Type:** trial · **Inclusion:** included

- **Intervention:** Datopotamab deruxtecan +/- durvalumab
- **Indication / model:** PD-L1 CPS >=10 locally recurrent inoperable or metastatic triple-negative breast cancer, 1L (1L); PD-L1 CPS >=10 by 22C3
- **Design:** randomized open-label phase 3 (TROPION-Breast05)
- **n:** 1075
- **Effect size:** — (readout pending) — PFS and OS
- **Case fit:** strong
- **Notes:** (toxicity table not extracted)

### Tolaney SM/— (2025)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** Sacituzumab tirumotecan (sac-TMT, MK-2870) 4 mg/kg IV q2w as monotherapy or + pembrolizumab
- **Indication / model:** Locally recurrent inoperable or metastatic TNBC, 1L, PD-L1 CPS <10 (1L); PD-L1 CPS <10 by 22C3
- **Design:** randomized open-label phase 3 (TroFuse-011)
- **n:** 1200
- **Effect size:** — (readout pending) — PFS / OS dual primary
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

### Tolaney SM/— (2025)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** Sacituzumab govitecan + pembrolizumab
- **Indication / model:** Triple-negative breast cancer with residual invasive disease after neoadjuvant therapy + surgery (post-neoadjuvant adjuvant) (adj); RD post KEYNOTE-522-style neoadjuvant
- **Design:** randomized open-label phase 3 (ASCENT-05 / OptimICE-RD / AFT-65)
- **n:** 1514
- **Effect size:** — (readout pending) — iDFS
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

## Tsai CJ/— (2025)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** SBRT to all sites of oligometastatic / oligoprogressive disease + ongoing systemic therapy
- **Indication / model:** Metastatic breast cancer with oligoprogressive or oligometastatic disease on systemic therapy (1L+ consolidation); oligometastatic or oligoprogressive (<=5 lesions)
- **Design:** randomized phase 2
- **n:** 200
- **Effect size:** — (ongoing) — PFS
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

## Tan TJ/Dent R (2024) — Clin Cancer Res

**Reference:**PMID 38236575 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Olaparib +/- durvalumab (maintenance)
- **Indication / model:** Platinum-pretreated advanced TNBC, maintenance after platinum response (maintenance); Advanced TNBC with SD/CR/PR on 1L-2L platinum chemo; HRD-positive / BRCA-mut enriched but not required; randomized 1:1.
- **Design:** Randomized open-label phase 2 (DORA, NCT03167619)
- **n:** 45
- **Effect size:** 35.7% (combo) vs 11.8% (olaparib alone) % — PFS at 12 months
- **Toxicities:** nausea (n=45, 35%); anemia (n=45, 31%); neutropenia G>=3 (n=45, 4.4%)
- **Case fit:** partial
- **Notes:** Chemo-free PARP-i + ICI maintenance after platinum induction; design template for a defensible off-trial regimen in this patient given her BRCA1 + TIL-rich + TMB-H phenotype. Requires prior platinum, so not 1L-enrollable.

## Bardia A/Rugo HS (2024) — J Clin Oncol

**Reference:**PMID 38422473 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Sacituzumab govitecan — ASCENT final analysis
- **Indication / model:** Pretreated metastatic TNBC (final database lock + HER2/TROP2 expression subgroups) (2L+)
- **Design:** Final analysis of randomized phase 3 (ASCENT)
- **n:** 529
- **Effect size:** 0.51 HR — Final OS analysis; HER2 / TROP2 expression subgroups
- **Variance:** p=<0.0001
- **Toxicities:** neutropenia G>=3 (n=258, 52%); treatment-related deaths G5 (n=258, 0.4%)
- **Case fit:** partial
- **Notes:** Final ASCENT readout confirms durability of SG OS benefit and shows efficacy is preserved across TROP2 expression strata — a useful counter to selection arguments for biomarker-restricted use.

## Tutt ANJ/Yap TA (2024)

**Reference:** — · **Type:** clinical · **Inclusion:** included

- **Intervention:** Olaparib + ceralasertib (ATR-i)
- **Indication / model:** Metastatic TNBC with HRR mutation stratification (2L+); mTNBC; stratified BRCA-mut vs HRR-mut non-BRCA vs HRR-WT; up to 2 prior chemo lines.
- **Design:** Randomized phase 2 (VIOLETTE, NCT03330847)
- **n:** 70
- **Effect size:** 0.81 HR — PFS by stratum
- **Variance:** p=not significant
- **Toxicities:** anemia G>=3 (n=23, 30%); thrombocytopenia G>=3 (n=23, 25%)
- **Case fit:** partial
- **Notes:** VIOLETTE is the dedicated PARPi + DDR combination phase 2 in mTNBC. Both combinations failed to beat olaparib monotherapy; adavosertib arm closed for toxicity. Decision-relevant evidence against routinely combining ATR/WEE1 with olaparib in BRCA1-mut TNBC outside of trials.

### Bardia A/Cortes J (2024) — J Clin Oncol

**Reference:**doi:10.1200/JCO.24.00920 · **Type:** clinical · **Inclusion:** excluded — TROPION-Breast01 enrolled HR+/HER2- breast cancer, not TNBC. Patient is HR-negative TNBC, so this trial's population does not match. The TROPION-Breast02 row covers the relevant 1L mTNBC indication for Dato-DXd.

- **Intervention:** Datopotamab deruxtecan (TROPION-Breast01)
- **Notes:** Excluded: TROPION-Breast01 enrolled HR+/HER2- breast cancer, not TNBC. Patient is HR-negative TNBC, so this trial's population does not match. The TROPION-Breast02 row covers the relevant 1L mTNBC indication for Dato-DXd. — TROPION-Breast01 was in HR+/HER2- mBC, OS at final analysis was not significantly improved (HR 1.01). Out of scope for this TNBC patient; covered by the TROPION-Breast02 row instead.

### Gruber JJ/Telli ML (2024)

**Reference:** — · **Type:** clinical · **Inclusion:** excluded — NCT03990896 (Stanford talazoparib in somatic BRCA1/2-mut mBC) is an ongoing single-arm phase 2 with only an interim n=13 cohort; no peer-reviewed primary publication of efficacy or safety data yet. Trial entry on the screener row is sufficient — clinical-evidence row would duplicate the trials.jsonl entry without adding a published-evidence base to summarize.

- **Intervention:** Talazoparib (somatic-BRCA expansion)
- **Notes:** Excluded: NCT03990896 (Stanford talazoparib in somatic BRCA1/2-mut mBC) is an ongoing single-arm phase 2 with only an interim n=13 cohort; no peer-reviewed primary publication of efficacy or safety data yet. Trial entry on the screener row is sufficient — clinical-evidence row would duplicate the trials.jsonl entry without adding a published-evidence base to summarize. — Trial-row only at this stage; no peer-reviewed clinical evidence to compile. Watch for primary publication.

### Jhaveri KL/Turner NC (2024) — N Engl J Med

**Reference:**PMID 39476340 · **Type:** clinical · **Inclusion:** excluded — INAVO120 (Jhaveri NEJM 2024, PMID 39476340) enrolled HR+/HER2- + PIK3CA-mut + endocrine-resistant breast cancer with inavolisib + palbociclib + fulvestrant. Indication is HR+, not TNBC; PFS HR 0.43 cannot be extrapolated to TNBC. The inavolisib + eribulin TNBC trial (NCT07441512) is already captured on the trials.jsonl screener row; no peer-reviewed TNBC inavolisib efficacy data exist yet.

- **Intervention:** Inavolisib (INAVO120, HR+ context)
- **Notes:** Excluded: INAVO120 (Jhaveri NEJM 2024, PMID 39476340) enrolled HR+/HER2- + PIK3CA-mut + endocrine-resistant breast cancer with inavolisib + palbociclib + fulvestrant. Indication is HR+, not TNBC; PFS HR 0.43 cannot be extrapolated to TNBC. The inavolisib + eribulin TNBC trial (NCT07441512) is already captured on the trials.jsonl screener row; no peer-reviewed TNBC inavolisib efficacy data exist yet. — Mechanism-of-action precedent in HR+; not directly applicable to TNBC and no TNBC efficacy data published. The pending inavolisib+eribulin trial is in the screener already.

## Schmid P/— (2024)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** Capivasertib 400 mg PO BID days 2-5 weekly + paclitaxel 80 mg/m^2 IV weekly
- **Indication / model:** Locally advanced or metastatic triple-negative breast cancer, 1L (1L); PIK3CA / AKT1 / PTEN alteration not required for enrollment but stratified
- **Design:** randomized double-blind placebo-controlled phase 3 (CAPItello-290)
- **n:** 800
- **Effect size:** — (primary readout reported negative for OS at 2024 ESMO) — OS in overall and AKT-pathway-altered populations
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

## NCI/— (2024)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** ZEN-3694 (BET inhibitor) + pembrolizumab + nab-paclitaxel
- **Indication / model:** Advanced TNBC, any line (2L+); MYC pathway not formally required for enrollment; rationale focused on basal/MYC-driven TNBC
- **Design:** single-arm phase 1b dose-finding
- **n:** 60
- **Effect size:** — (early-phase, MYC-amplified TNBC subgroup analysis preplanned) — MTD / RP2D, then ORR
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

## Gruber JJ/Telli ML (2024)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** Talazoparib 1 mg PO daily
- **Indication / model:** Metastatic breast cancer with somatic BRCA1/2 mutation (1L-3L); somatic (non-germline) BRCA1 or BRCA2 mutation
- **Design:** single-arm phase 2
- **n:** 13
- **Effect size:** ORR ~38% in interim analysis — ORR
- **Case fit:** strong

- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

### Tutt ANJ/— (2024)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** Olaparib + ceralasertib (AZD6738) vs olaparib + adavosertib (closed) vs olaparib monotherapy
- **Indication / model:** Metastatic TNBC stratified by HRR mutation status (2L+); HRR mutation including BRCA1/2 (stratified); HR-deficient enriched
- **Design:** randomized phase 2 (VIOLETTE)
- **n:** 70
- **Effect size:** Olaparib + ceralasertib PFS HR ~0.81 in BRCA-mut; ~0.93 in HRR-mut non-BRCA — PFS
- **Variance:** p=not significant
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

### —/— (2023)

**Reference:** — · **Type:** clinical · **Inclusion:** excluded — EPIK-B3 (NCT04251533) — phase 3 alpelisib + nab-paclitaxel in PIK3CA-mut or PTEN-loss mTNBC was terminated early due to slow recruitment and failure of Part B1 to meet the primary ORR endpoint. No peer-reviewed publication of efficacy data; not enrollable; the most directly TNBC-relevant alpelisib trial returned a negative readout, which removes alpelisib from the practical option set for this patient even though her PIK3CA-mut + PTEN-loss profile would have been the targeted subgroup.

- **Intervention:** Alpelisib (EPIK-B3)
- **Notes:** Excluded: EPIK-B3 (NCT04251533) — phase 3 alpelisib + nab-paclitaxel in PIK3CA-mut or PTEN-loss mTNBC was terminated early due to slow recruitment and failure of Part B1 to meet the primary ORR endpoint. No peer-reviewed publication of efficacy data; not enrollable; the most directly TNBC-relevant alpelisib trial returned a negative readout, which removes alpelisib from the practical option set for this patient even though her PIK3CA-mut + PTEN-loss profile would have been the targeted subgroup. — Part A recruitment halted Nov 2022; Part B1 missed primary ORR; Part B2 never initiated. No published efficacy paper to extract — only the clinicaltrials.gov record and sponsor statement. Decision-relevant only as evidence against pursuing alpelisib in TNBC even with a hotspot PIK3CA call.

### —/— (2023)

**Reference:** — · **Type:** clinical · **Inclusion:** excluded — Samuraciclib (CT7001) phase 1/2 TNBC monotherapy and combination data exist only as conference abstracts (2021-2023) with very small TNBC cohorts (<20 evaluable patients); no peer-reviewed primary publication of efficacy in TNBC. The mechanistic rationale for MYC-amp targeting is real but the published evidence base is too thin to support a row beyond abstract-only summarization.

- **Intervention:** Samuraciclib (CDK7 inhibitor)
- **Notes:** Excluded: Samuraciclib (CT7001) phase 1/2 TNBC monotherapy and combination data exist only as conference abstracts (2021-2023) with very small TNBC cohorts (<20 evaluable patients); no peer-reviewed primary publication of efficacy in TNBC. The mechanistic rationale for MYC-amp targeting is real but the published evidence base is too thin to support a row beyond abstract-only summarization. — Abstract-only data; full TNBC publication pending. CDK7/MYC-amp mechanism remains hypothesis-generating only.

### Geyer CE Jr/Loibl S (2022) — Ann Oncol

**Reference:**PMID 35093516 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Veliparib + carboplatin + paclitaxel (BrighTNess 4-year follow-up)
- **Indication / model:** Stage II-III triple-negative breast cancer, neoadjuvant (neoadj); Same 634-patient cohort, 4.5-yr median follow-up
- **Design:** Randomized phase 3 long-term follow-up (BrighTNess)
- **n:** 634
- **Effect size:** 0.63 HR — EFS at 4-year follow-up
- **Variance:** 95% CI 0.43–0.92; p=0.02
- **Toxicities:** second primary malignancy (incl MDS/AML) (n=628, 1.6%)
- **Case fit:** weak
- **Notes:** Long-term follow-up confirms that adding carboplatin (not veliparib) drives the EFS gain. Decision-relevant for choosing platinum-containing 1L systemic backbone in a BRCA-mut patient.

### Cortes J/Schmid P (2022) — N Engl J Med

**Reference:**PMID 35857659 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Pembrolizumab + chemotherapy (KEYNOTE-355 final OS)
- **Indication / model:** Previously untreated mTNBC, PD-L1 CPS >=10 final OS (1L); Same 847-patient cohort; final OS analysis at 44-mo median follow-up.
- **Design:** Final OS analysis, randomized phase 3 (KEYNOTE-355)
- **n:** 847
- **Effect size:** 0.73 HR — OS in PD-L1 CPS >=10
- **Variance:** 95% CI 0.55–0.95; p=0.0185
- **Toxicities:** any treatment-related AE G>=3 (n=562, 68.1%); treatment-related deaths G5 (n=562, 0.4%)
- **Case fit:** strong
- **Notes:** Final OS readout confirming 7-mo median OS gain in CPS >=10. The defining survival evidence for 1L pembrolizumab + chemo in mTNBC.

### Khan SA/Wolff AC (2022) — J Clin Oncol

**Reference:**PMID 34995128 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Early locoregional therapy of intact primary in de novo stage IV breast cancer
- **Indication / model:** De novo stage IV breast cancer with intact primary tumor, no progression after 4-8 mo of optimal systemic therapy (1L+ consolidation); Stage IV intact primary; 50% HR+/HER2-, 29% HER2+, 10% TNBC, 11% other; bone-only 31%, viscera-only 26%, both 27%; median age 55.
- **Design:** Randomized phase 3 (ECOG-ACRIN E2108)
- **n:** 256
- **Effect size:** 1.11 HR — Overall survival
- **Variance:** 95% CI 0.82–1.52; p=0.57
- **Toxicities:** surgical complications (n=125, 15%)
- **Case fit:** strong
- **Notes:** Definitive RCT showing no OS benefit from resecting an intact primary in de novo M1 breast cancer. The patient's locoregional control may still warrant local therapy if symptomatic — the trial does not say no, it

says no OS benefit. Decision-relevant counter-evidence to routine surgical consolidation.

### Aftimos PG/Schmid P (2022)

**Reference:** — · **Type:** clinical · **Inclusion:** included

- **Intervention:** ZEN-3694 (BET inhibitor) + talazoparib
- **Indication / model:** Metastatic TNBC without germline BRCA1/2 mutations (2L+); mTNBC; germline BRCA1/2-WT required; up to 3 prior lines; MYC-driven / basal-like enriched.
- **Design:** Phase 1b/2 single-arm (NCT03901469)
- **n:** 51
- **Effect size:** 22 % — ORR
- **Toxicities:** thrombocytopenia G>=3 (n=51, 30%); anemia G>=3 (n=51, 18%); fatigue (n=51, 40%)
- **Case fit:** weak
- **Notes:** ASCO 2022 abstract only; full publication pending. The trial restricted to BRCA-WT TNBC, so it does NOT directly apply to this patient (BRCA1-mut). Included to anchor the BET-inhibitor evidence base for the MYC-amplification axis — patient is BRCA1-mut so this regimen is not enrollable for her.

### Modi S/Harbeck N (2022) — N Engl J Med

**Reference:**PMID 35665782 · **Type:** clinical · **Inclusion:** excluded — Patient is HER2 fully negative (HER2 0 by IHC implied by 'HER2-negative' on profile); DESTINY-Breast04 enrolled HER2-LOW (IHC 1+ or 2+/FISH-) — patient does not qualify unless reflex IHC confirms HER2-low status. Per profile.json the HER2-0 vs HER2-low distinction is not specified; flagged as not applicable pending confirmation.

- **Intervention:** Trastuzumab deruxtecan
- **Notes:** Excluded: Patient is HER2 fully negative (HER2 0 by IHC implied by 'HER2-negative' on profile); DESTINY-Breast04 enrolled HER2-LOW (IHC 1+ or 2+/FISH-) — patient does not qualify unless reflex IHC confirms HER2-low status. Per profile.json the HER2-0 vs HER2-low distinction is not specified; flagged as not applicable pending confirmation. — DESTINY-Breast04 established T-DXd in HER2-low mBC. Patient documented HER2 'negative' without specifying IHC 0 vs 1+/2+. If reflex confirms HER2-low (1+ or 2+/FISH-), T-DXd becomes an option; if HER2-0, the regimen is not labeled. Flagged as not applicable pending HER2 IHC clarification.

### Lloyd/O'Connor (2022) — Molecular Cancer Therapeutics

**Reference:**PMID 35046096 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Ceralasertib (AZD6738, ATR inhibitor) + olaparib
- **Indication / model:** BRCA2-mutant TNBC PDX and BRCA-wild-type TNBC PDX; ATM-deficient lung and breast cell lines
- **Design:** ATR inhibition blocks the BRCA1-independent RAD51 loading and replication-fork protection that drive PARPi resistance, while sustaining the replication-stress lethality that PARP trapping initiates
- **n:** n=8-10 mice/arm per PDX
- **Effect size:** strong — Complete tumour regression of a BRCA2-mutant TNBC PDX with intermittent ceralasertib plus continuous olaparib; in BRCA-wild-type TNBC PDX, intensified ceralasertib dosing still produced regression. The preclinical rationale that drove VIOLETTE and the ATR+PARP combination trial space.
- **Variance:** vs Vehicle, olaparib monotherapy, ceralasertib monotherapy; translatability: high

- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** VIOLETTE clinical readout was negative for the combination over olaparib alone — the preclinical synergy is robust, but tolerability of intermittent ATRi at the active dose has been a clinical headwind.

### DiMarco/Roberts (2022) — Cell Reports

**Reference:**PMID 35168954 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** APOBEC mutagenesis as an immune-response amplifier (rationale for ICI in TMB-H APOBEC-rich TNBC)
- **Indication / model:** Syngeneic mouse breast tumor models with engineered APOBEC3B expression; matched RNA-seq immune deconvolution
- **Design:** APOBEC-induced cytidine deamination generates a high-affinity neoantigen pool and triggers IFN-γ-dependent T-cell infiltration; the resulting immune contracture both restrains tumor growth and primes responsiveness to PD-1 blockade
- **n:** n=8-10 mice/arm across 3 tumor lines
- **Effect size:** moderate — APOBEC3B-expressing syngeneic breast tumors grew slower than APOBEC-null controls in immunocompetent hosts but identically in T-cell-depleted mice, and they were more responsive to anti-PD-1. The mechanistic preclinical case that an APOBEC-signature, TMB-H, TIL-rich TNBC (exactly this patient's phenotype) is an ICI-responsive subset.
- **Variance:** vs APOBEC3B-null isogenic tumors; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Syngeneic mouse engineering rather than patient-derived models; the human-translation link is correlative (TMB / APOBEC-sig / IFN-γ score). Supports the rationale to anchor the TMB-14 + TIL-3+ signal as ICI-favorable.

### Tutt ANJ/Geyer CE Jr (2021) — N Engl J Med

**Reference:**PMID 34081848 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Olaparib (adjuvant)
- **Indication / model:** High-risk early-stage HER2-negative breast cancer with germline BRCA1/2 mutation (adjuvant) (adj); Germline BRCA1/2 carriers; high-risk early breast cancer (TNBC or HR+); residual disease after neoadjuvant chemo OR pT2/pN1 after adjuvant chemo.
- **Design:** Randomized double-blind placebo-controlled phase 3 (OlympiA)
- **n:** 1836
- **Effect size:** 0.58 HR — iDFS
- **Variance:** 95% CI 0.46–0.74; p=<0.001
- **Toxicities:** any treatment-related AE G>=3 (n=911, 24.5%); anemia G>=3 (n=911, 8.7%); neutropenia G>=3 (n=911, 4.8%); treatment discontinuation due to AE (n=911, 9.9%); MDS/AML (n=911, 0.2%)
- **Case fit:** weak
- **Notes:** OlympiA is the adjuvant standard for germline-BRCA early breast cancer. Patient is de novo M1, not adjuvant, so the label does not apply. Included for evidence-base completeness on the BRCA1 PARP-inhibitor axis; supports post-consolidation olaparib if oligometastatic conversion is pursued.

## Turner NC/Cameron DA (2021) — Clin Cancer Res

**Reference:**PMID 34301749 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Niraparib
- **Indication / model:** Germline BRCA1/2-mutated HER2-negative advanced breast cancer (1L-3L); Germline BRCA1/2 carriers; HER2-neg locally advanced or metastatic; up to 2 prior chemo lines; ECOG 0-2.
- **Design:** Randomized open-label phase 3 (BRAVO, EORTC 1307-BCG)
- **n:** 215
- **Effect size:** 0.96 HR — PFS (central review)
- **Variance:** 95% CI 0.65–1.44; p=not significant
- **Toxicities:** thrombocytopenia G>=3 (n=141, 28.9%); anemia G>=3 (n=141, 24.8%); neutropenia G>=3 (n=141, 19.9%); treatment discontinuation due to AE (n=141, 14.9%)
- **Case fit:** partial
- **Notes:** Trial halted for futility on central review, though local PFS suggested benefit — the central-vs-local discordance is the published interpretation caveat. Niraparib is not FDA-approved in breast cancer; off-label only and likely to face payer pushback vs labeled olaparib/talazoparib.

## Bardia A/Rugo HS (2021) — N Engl J Med

**Reference:**PMID 33882206 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Sacituzumab govitecan
- **Indication / model:** Pretreated metastatic triple-negative breast cancer (>=2 prior lines including a taxane) (2L+); mTNBC after >=2 prior lines in advanced setting, must include a taxane; brain-mets cohort separate.
- **Design:** Randomized open-label phase 3 (ASCENT)
- **n:** 529
- **Effect size:** 0.41 HR — PFS in brain-mets-negative cohort (BICR)
- **Variance:** 95% CI 0.32–0.52; p=<0.001
- **Toxicities:** neutropenia G>=3 (n=258, 51%); diarrhea G>=3 (n=258, 10%); anemia G>=3 (n=258, 8%); febrile neutropenia G>=3 (n=258, 6%); treatment discontinuation due to AE (n=258, 4.7%)
- **Case fit:** partial
- **Notes:** Pivotal RCT supporting FDA approval of sacituzumab govitecan in 2L+ mTNBC. Patient is 1L-treatment-naive so not directly enrollable; anchors the basal-like/TROP2-ADC axis for the likely 2L option after progression on 1L ICI + chemo.

## Miles D/Gianni L (2021) — Ann Oncol

**Reference:**PMID 34219000 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Atezolizumab + paclitaxel (IMpassion131)
- **Indication / model:** Previously untreated metastatic TNBC (1L); 1L mTNBC; randomized 2:1 atezo+paclitaxel vs placebo+paclitaxel; 45% PD-L1+.
- **Design:** Randomized double-blind placebo-controlled phase 3 (IMpassion131)
- **n:** 651
- **Effect size:** 0.82 HR — PFS in PD-L1+ subgroup (primary)
- **Variance:** 95% CI 0.6–1.12; p=0.20
- **Toxicities:** any treatment-related AE G>=3 (n=431, 49%)

- **Case fit:** weak
- **Notes:** Negative confirmatory trial: paclitaxel solvent (vs nab-paclitaxel) is the leading published hypothesis for the IMpassion130/131 discordance. The failure triggered the US withdrawal of atezolizumab in mTNBC. Included as the evidence anchor for why the patient should be steered to the pembrolizumab + chemo backbone rather than atezolizumab.

### Keenan TE/Tolaney SM (2021) — Clin Cancer Res

**Reference:**PMID 33257427 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Adavosertib + cisplatin
- **Indication / model:** Metastatic triple-negative breast cancer, 1L-2L (1L-2L); mTNBC; 1L or 2L; BRCA-mut and BRCA-WT both eligible; TP53 status not stratified.
- **Design:** Single-arm phase 2 (DFCI-Tolaney, NCT03012477)
- **n:** 34
- **Effect size:** 26 % — ORR
- **Variance:** 95% CI 13–44
- **Toxicities:** any treatment-related AE G>=3 (n=34, 53%); diarrhea G>=3 (n=34, 21%); neutropenia G>=3 (n=34, 15%); anemia G>=3 (n=34, 12%)
- **Case fit:** partial
- **Notes:** Adavosertib + cisplatin missed the pre-specified 30% ORR threshold in mTNBC and no genomic correlate (including TP53) predicted response. Tempers the WEE1-inhibitor rationale for TP53-mut TNBC; remains experimental, trial-only.

### Chmura SJ/White JR (2021) — JAMA Oncol

**Reference:**PMID 33885704 · **Type:** clinical · **Inclusion:** included

- **Intervention:** SBRT or surgical resection of oligometastatic breast cancer (NRG-BR002)
- **Indication / model:** Newly oligometastatic breast cancer (1-4 metastases) of any subtype, on standard systemic therapy (1L+ consolidation); Breast-specific cohort; 1-4 oligometastases; on standard systemic therapy; randomized to add consolidation vs continue systemic alone.
- **Design:** Randomized phase IIR/III (NRG-BR002)
- **n:** 125
- **Effect size:** 1.31 HR — Safety (primary); PFS phase IIR readout negative at SABCS 2024
- **Variance:** p=ns
- **Toxicities:** any treatment-related AE G>=3 (n=65, 12%)
- **Case fit:** strong
- **Notes:** Breast-specific oligometastatic consolidation trial — phase IIR PFS readout was negative for consolidation (HR ~1.31), tempering the curative-intent framing for this de novo oligometastatic patient. Decision-relevant evidence against routine SBRT/resection over systemic therapy alone; biology-driven case-by-case board discussions still continue.

### Okajima/Agatsuma (2021) — Molecular Cancer Therapeutics

**Reference:**PMID 34413126 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Datopotamab deruxtecan (Dato-DXd, DS-1062)

- **Indication / model:** TROP2-expressing NSCLC and breast cell-line xenografts; admixed TROP2+/TROP2- xenografts for bystander testing
- **Design:** Humanized anti-TROP2 IgG1 conjugated to the exatecan-derivative DXd via a cleavable tetrapeptide linker; high membrane permeability of the released DXd payload drives a strong bystander effect on neighbouring TROP2-low cells
- **n:** n=5-8 mice/arm across multiple cell-line xenografts
- **Effect size:** strong — Dato-DXd produced regressions in TROP2-expressing xenografts at 3 mg/kg q3w; the admixed-cell model confirmed bystander killing of TROP2-negative tumor cells when juxtaposed with TROP2-positive cells. The discovery preclinical that underpins TROPION-Breast02 and TROPION-Breast05.
- **Variance:** vs Vehicle; isotype-control ADC; trastuzumab deruxtecan as the platform comparator; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** TNBC-specific PDX work is limited in the discovery paper; subsequent IDEAL and gynae-tumor extensions filled in the basal-like context.

## Schmid/Saura (2021) — Breast Cancer Research

**Reference:**PMID 33451345 · **Type:** preclinical · **Inclusion:** excluded — Reviewed for the IRS2-amplification rationale but the available xentuzumab preclinical literature does not match the patient's IRS2-amplified TNBC context — published xentuzumab preclinical data centers on IGF-1/IGF-2 ligand neutralisation in HR+ and NSCLC models, with no published xenograft data in an IRS2-amplified TNBC background. The xentuzumab Phase II HR+ readout was negative and the program was discontinued, further weakening the case for compiling a preclinical row.

- **Intervention:** Xentuzumab (BI 836845, IGF-1/IGF-2 neutralising antibody)
- **Indication / model:** Phase Ib/II clinical trial in HR+/HER2- metastatic breast cancer (not a preclinical paper)
- **n:** Clinical n=140 patients
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Reviewed for the IRS2-amplification rationale but the available xentuzumab preclinical literature does not match the patient's IRS2-amplified TNBC context — published xentuzumab preclinical data centers on IGF-1/IGF-2 ligand neutralisation in HR+ and NSCLC models, with no published xenograft data in an IRS2-amplified TNBC background. The xentuzumab Phase II HR+ readout was negative and the program was discontinued, further weakening the case for compiling a preclinical row. — IRS2-axis preclinical evidence in TNBC is genuinely thin in the published literature; IGF-1R-axis approaches sit in trial-territory without a strong preclinical anchor.

## Diéras V/Arun BK (2020) — Lancet Oncol

**Reference:**PMID 32861273 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Veliparib + carboplatin + paclitaxel
- **Indication / model:** HER2-negative advanced breast cancer with germline BRCA1/2 mutation (1L-3L); Germline BRCA1/2 carriers; HER2-neg locally advanced or metastatic breast cancer; ECOG 0-2; up to 2 prior chemo lines, including 1 prior platinum allowed if 6+ months prior.
- **Design:** Randomized double-blind placebo-controlled phase 3 (BROCADE3)
- **n:** 509
- **Effect size:** 0.71 HR — PFS

- **Variance:** 95% CI 0.57–0.88; p=0.0016
- **Toxicities:** neutropenia G>=3 (n=337, 81%); anemia G>=3 (n=337, 42%); thrombocytopenia G>=3 (n=337, 40%); treatment discontinuation due to AE (n=337, 7.7%)
- **Case fit:** strong
- **Notes:** PFS improvement did not translate into OS benefit at final analysis. Veliparib was never FDA-approved in breast cancer; the regimen anchors the BRCA-mut platinum/PARP-i combination concept but is not commercially available. Useful as evidence base; not directly prescribable.

### Domchek SM/Kaufman B (2020) — Lancet Oncol

**Reference:**PMID 32771088 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Olaparib + durvalumab
- **Indication / model:** Germline BRCA-mutated HER2-negative metastatic breast cancer (1L-3L); Germline BRCA1/2 carriers; HER2-neg mBC; up to 2 prior chemo lines; ECOG 0-1.
- **Design:** Single-arm phase 1/2 basket expansion (MEDIOLA breast cohort)
- **n:** 34
- **Effect size:** 80 % — DCR at 12 weeks
- **Variance:** 95% CI 64.3–90.9
- **Toxicities:** any treatment-related AE G>=3 (11/34, 32%); anemia G>=3 (n=34, 12%); neutropenia G>=3 (n=34, 9%); pancreatitis G>=3 (n=34, 6%); treatment discontinuation due to AE (3/34)
- **Case fit:** strong
- **Notes:** Small but mechanistically clean PARP-i + PD-L1 cohort in germline-BRCA HER2-neg mBC. Supports an olaparib + durvalumab maintenance strategy after platinum-containing 1L induction; regimen is off-label outside of trials.

### Cortes J/Schmid P (2020) — Lancet

**Reference:**PMID 33278935 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Pembrolizumab + chemotherapy (1L mTNBC)
- **Indication / model:** Previously untreated locally recurrent inoperable or metastatic TNBC (1L); 1L mTNBC; PD-L1 stratified by 22C3 CPS; ECOG 0-1; primary endpoint in CPS >=10 subgroup.
- **Design:** Randomized double-blind placebo-controlled phase 3 (KEYNOTE-355)
- **n:** 847
- **Effect size:** 0.65 HR — PFS in CPS >=10 (primary)
- **Variance:** 95% CI 0.49–0.86; p=0.0012
- **Toxicities:** any treatment-related AE G>=3 (n=562, 68.1%); neutropenia G>=3 (n=562, 41%); immune-mediated AE G>=3 (n=562, 5.3%); treatment-related deaths G5 (n=562, 0.4%)
- **Case fit:** strong
- **Notes:** Pivotal trial driving FDA approval of pembrolizumab + chemo for 1L mTNBC CPS >=10. Patient is treatment-naive with TIL-rich and TMB-H phenotype; PD-L1 CPS is pending and gates the regimen.

### Schmid P/O'Shaughnessy J (2020) — N Engl J Med

**Reference:**PMID 32101663 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Pembrolizumab + neoadjuvant chemo (KEYNOTE-522)

- **Indication / model:** Stage II-III triple-negative breast cancer, neoadjuvant (neoadj); Stage II-III TNBC; PD-L1-agnostic; ECOG 0-1.
- **Design:** Randomized double-blind placebo-controlled phase 3 (KEYNOTE-522)
- **n:** 1174
- **Effect size:** pCR 64.8% vs 51.2%; EFS HR 0.63 % — pCR (primary); EFS at later analysis
- **Variance:** 95% CI 0.48–0.82; p=<0.001
- **Toxicities:** any treatment-related AE G>=3 (n=784, 78%); immune-mediated AE G>=3 (n=784, 14%); endocrine disorders (n=784, 23%)
- **Case fit:** weak
- **Notes:** Neoadjuvant TNBC trial; the patient is M1 not stage II-III so the regimen is not directly applicable. Included because the TIL-3+/TMB-14 phenotype matches the subgroup with the largest pCR gain, which reinforces the case for a KEYNOTE-355-style 1L ICI+chemo regimen in metastatic disease.

## Marabelle A/Bang YJ (2020) — Lancet Oncol

**Reference:**PMID 32919526 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Pembrolizumab monotherapy in TMB-H solid tumors
- **Indication / model:** Advanced solid tumors with TMB-H (>=10 mut/Mb by FoundationOne CDx) (2L+); Prospectively-defined TMB-H cohort across 10 tumor types; no prior anti-PD-(L)1; failed >=1 prior line; included anal, biliary, cervical, endometrial, mesothelioma, neuroendocrine, salivary, small-cell lung, thyroid, vulvar — breast was NOT in the basket.
- **Design:** Single-arm, prospectively-defined TMB-H cohort within multi-cohort phase 2 (KEYNOTE-158)
- **n:** 102
- **Effect size:** 29 % — ORR
- **Variance:** 95% CI 21–39
- **Toxicities:** any treatment-related AE G>=3 (n=105, 15%); colitis G>=3 (2/105); pneumonitis G5 (1/105)
- **Case fit:** cross_tumor_only
- **Notes:** KEYNOTE-158 supports the tumor-agnostic pembrolizumab indication for TMB-H >=10 mut/Mb on FoundationOne CDx. Breast cancer was NOT in the original basket — application to this patient is cross-tumor extrapolation. Label requires prior progression and no satisfactory alternatives; not yet applicable in 1L.

## Pitts/Pitts (2020) — Frontiers in Oncology

**Reference:**PMID 32204315 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Adavosertib (WEE1 inhibitor) + capecitabine/5-FU
- **Indication / model:** TNBC cell line panel (MDA-MB-468, HCC1937, CAL-51, MDA-MB-231) and TNBC PDX models; PDX of basal-like TNBC
- **Design:** WEE1 inhibition compounds the replication stress from thymidylate synthase inhibition by 5-FU, driving p53-null TNBC cells through unrepaired S-phase damage
- **n:** n=6-10 mice/arm across 3 PDX models
- **Effect size:** moderate — Synergistic combination effects in MDA-MB-468, HCC1937 and CAL-51 cell lines; PDX cohorts showed enhanced tumour growth inhibition with the combination over either single agent. HCC1937 is the BRCA1-mutant basal-like TNBC line most directly relevant to this patient.
- **Variance:** vs Vehicle, adavosertib monotherapy, capecitabine monotherapy; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong

- **Notes:** Capecitabine is a standard later-line TNBC backbone; the rationale fits a 2L+ context rather than the current 1L decision.

### Vinayak S/Telli ML (2019) — JAMA Oncol

**Reference:**PMID 31194225 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Niraparib + pembrolizumab
- **Indication / model:** Advanced or metastatic triple-negative breast cancer (2L+); Advanced TNBC; BRCA-mut (n=15) and BRCA-WT (n=27) both eligible; up to 2 prior cytotoxic lines.
- **Design:** Single-arm phase 1/2 (TOPACIO/KEYNOTE-162)
- **n:** 55
- **Effect size:** 21 % — ORR
- **Variance:** 95% CI 12–33
- **Toxicities:** anemia G>=3 (10/55, 18%); thrombocytopenia G>=3 (8/55, 15%); fatigue G>=3 (4/55, 7%); immune-related AE (any) (n=55, 15%); immune-related AE G>=3 (n=55, 4%)
- **Case fit:** strong
- **Notes:** The 47% ORR in BRCA-mut TNBC is the foundational chemo-free PARP-i + ICI signal. Patient's BRCA1 + TIL-rich + TMB-H profile is the canonical TOPACIO responder phenotype; niraparib off-label in breast cancer.

### Palma DA/Senan S (2019) — Lancet

**Reference:**PMID 30982687 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Stereotactic ablative radiotherapy (SABR) for oligometastatic disease
- **Indication / model:** Controlled primary solid tumor with 1-5 metastatic lesions (mixed histologies, including breast) (any); Multiple primary sites; 18% breast; 1-5 oligometastases; controlled primary.
- **Design:** Randomized phase 2 open-label (SABR-COMET)
- **n:** 99
- **Effect size:** 0.57 HR — OS
- **Variance:** 95% CI 0.3–1.1; p=0.090
- **Toxicities:** any treatment-related AE G>=2 (n=66, 29%); treatment-related deaths G5 (3/66)
- **Case fit:** cross_tumor_only
- **Notes:** Foundational randomized evidence for SABR consolidation in mixed-tumor oligometastases. Patient is breast-only-liver oligomet (single 1.5 cm lesion); SABR-COMET is a small phase 2 across multiple tumor types — the OS HR is encouraging but NRG-BR002 in breast-specific cohort tempers extrapolation.

### Litton JK/Blum JL (2018) — N Engl J Med

**Reference:**PMID 30110579 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Talazoparib
- **Indication / model:** Locally advanced or metastatic HER2-negative breast cancer with germline BRCA1/2 mutation (1L-3L); Germline BRCA1/2 carriers; 45% TNBC, 55% HR+; ECOG 0-2; prior taxane or anthracycline required unless contraindicated; up to 3 prior chemo lines.
- **Design:** Randomized open-label phase 3 (EMBRACA)
- **n:** 431
- **Effect size:** 0.54 HR — PFS (BICR)

- **Variance:** 95% CI 0.41–0.71; p=<0.001
- **Toxicities:** anemia G>=3 (n=286, 39%); neutropenia G>=3 (n=286, 21%); thrombocytopenia G>=3 (n=286, 15%); fatigue (n=286, 50.3%); treatment discontinuation due to AE (n=286, 7.7%)
- **Case fit:** strong
- **Notes:** Pivotal registrational RCT for talazoparib in germline-BRCA HER2-negative mBC. Choice between olaparib and talazoparib usually rests on hematologic-toxicity tolerance — talazoparib has substantially higher anemia rates.

### Loibl S/Geyer CE Jr (2018) — Lancet Oncol

**Reference:**PMID 29501363 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Veliparib + carboplatin + paclitaxel (neoadjuvant)
- **Indication / model:** Stage II-III triple-negative breast cancer, neoadjuvant (neoadj); Treatment-naive stage II-III TNBC; BRCA-mut and BRCA-WT both eligible; 3-arm trial.
- **Design:** Randomized double-blind placebo-controlled phase 3 (BrighTNess)
- **n:** 634
- **Effect size:** 53% (Arm A) vs 58% (Arm B) vs 31% (Arm C) % — pCR (ypT0/is ypN0)
- **Variance:** p=<0.0001 (Arm A vs C); 0.36 (Arm A vs B)
- **Toxicities:** neutropenia G>=3 (n=628, 56%); anemia G>=3 (n=628, 29%); thrombocytopenia G>=3 (n=628, 12%)
- **Case fit:** weak
- **Notes:** Patient is M1 not stage II-III, so neoadjuvant regimen does not directly apply. Included because BrighTNess shows the platinum component drives the pCR gain, not veliparib — informs how to weight platinum vs PARP-i in the 1L systemic regimen for this BRCA1 patient.

### Tutt A/Bliss JM (2018) — Nat Med

**Reference:**PMID 29713086 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Carboplatin (vs docetaxel) in BRCA-mutated mTNBC — TNT trial
- **Indication / model:** Metastatic or recurrent locally advanced TNBC or BRCA1/2-mutant breast cancer (1L-2L); Unselected advanced TNBC (376 pts); 43 patients with germline BRCA1/2 mutations identified by prospective biomarker analysis.
- **Design:** Randomized open-label phase 3 (TNT, CRUK/07/012)
- **n:** 376
- **Effect size:** BRCA-mut: 68.0% (carbo) vs 33.3% (doce); ORR interaction p=0.01 % — ORR overall and in BRCA-mut subgroup
- **Variance:** p=0.01 (biomarker x treatment interaction)
- **Toxicities:** neutropenia G>=3 (n=188, 12%); thrombocytopenia G>=3 (n=188, 7%); neuropathy G>=3 (n=188, 11%)
- **Case fit:** strong
- **Notes:** TNT is the platinum-vs-taxane RCT establishing that BRCA-mut TNBC has double the carboplatin ORR vs docetaxel (68% vs 33%) — the foundational platinum-sensitivity data in BRCA1-mut TNBC. Anchors a carboplatin-containing 1L backbone for this patient.

### Schmid P/Emens LA (2018) — N Engl J Med

**Reference:**PMID 30345906 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Atezolizumab + nab-paclitaxel
- **Indication / model:** Previously untreated metastatic TNBC (1L); 1L mTNBC; primary endpoints in ITT and PD-L1 IC >=1% by SP142.
- **Design:** Randomized double-blind placebo-controlled phase 3 (IMpassion130)
- **n:** 902
- **Effect size:** 0.62 HR — PFS overall and in PD-L1+ subgroup
- **Variance:** 95% CI 0.49–0.78; p=<0.001
- **Toxicities:** treatment discontinuation due to AE (n=452, 15.9%); hypothyroidism (n=452, 17.3%); neutropenia G>=3 (n=452, 8%)
- **Case fit:** weak
- **Notes:** IMpassion130 led to the original accelerated approval of atezo+nab-paclitaxel in PD-L1 SP142 IC >=1% mTNBC; the indication was voluntarily withdrawn in 2021 after IMpassion131 failed to confirm benefit. Kept as historical anchor — the patient cannot access this regimen in the US.

### Cruz/Serra (2018) — Annals of Oncology

**Reference:**PMID 29635390 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Olaparib / talazoparib (functional HRD biomarker)
- **Indication / model:** BRCA1- and BRCA2-mutant breast cancer patient-derived xenografts (PDX) and matched germline-BRCA mBC tumor samples
- **Design:** Quantitative nuclear RAD51 foci in cycling tumor cells report functional HR competence — independent of the upstream BRCA1 lesion type
- **n:** PDX cohort: 18 BRCA1/2-mutated breast PDX models, plus 33 matched germline-BRCA patient samples
- **Effect size:** moderate — Low RAD51 foci predicted olaparib response in BRCA1/2-mutant PDX and patient tumours; restored RAD51 was the unifying feature of acquired and primary PARPi resistance regardless of whether resistance came from reversion, 53BP1 loss, or fork-protection rescue. The functional assay outperforms genomic-scar metrics in models where the scar is locked in but HR has rebounded.
- **Variance:** vs Treatment-naive vs olaparib-resistant tumours, with isogenic comparisons in selected models; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** RAD51 foci assay requires fresh-frozen or carefully-fixed tissue and cycling cells; not yet a CLIA-routine test.

### Patel/Coombes (2018) — Molecular Cancer Therapeutics

**Reference:**PMID 29545334 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Samuraciclib (CT7001 / ICEC0942, oral CDK7 inhibitor)
- **Indication / model:** Breast, colon and AML cell-line panel including TNBC lines (MDA-MB-468); MCF-7 xenografts as the primary in-vivo model
- **Design:** ATP-competitive selective CDK7 inhibition (IC50 40 nM, 15-230x selectivity over CDK1/2/5/9); shuts down RNA Pol II initiation phosphorylation and CDK-activating kinase function
- **n:** n=12 mice/arm in MCF-7 xenografts
- **Effect size:** moderate — 60% tumour growth inhibition in MCF-7 xenografts at 100 mg/kg with reduced ER

S118 phosphorylation; activity reported in TNBC models in the same paper. The discovery preclinical for the molecule now in clinical development as samuraciclib in MYC-driven TNBC.
- **Variance:** vs Vehicle; tamoxifen monotherapy; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** The primary in-vivo readout is MCF-7 (ER+) rather than a basal/MYC-amplified TNBC model. TNBC translation is mostly in-vitro within this paper.

### Robson M/Conte P (2017) — N Engl J Med

**Reference:**PMID 28578601 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Olaparib
- **Indication / model:** HER2-negative metastatic breast cancer with germline BRCA1/2 mutation (TNBC or HR+; up to 2 prior chemo lines) (1L-3L); Germline BRCA1/2 carriers; 50% TNBC, 50% HR+; ECOG 0-1; prior anthracycline and taxane required; up to 2 prior chemotherapy lines for metastatic disease.
- **Design:** Randomized open-label phase 3 (OlympiAD)
- **n:** 302
- **Effect size:** 0.58 HR — PFS (BICR)
- **Variance:** 95% CI 0.43–0.8; p=<0.001
- **Toxicities:** any treatment-related AE G>=3 (n=205, 36.6%); anemia G>=3 (n=205, 16.1%); neutropenia G>=3 (n=205, 9.3%); fatigue (n=205, 28.8%); nausea (n=205, 58%); treatment discontinuation due to AE (n=205, 4.9%)
- **Case fit:** strong
- **Notes:** Pivotal registrational RCT supporting FDA approval of olaparib in germline-BRCA HER2-negative mBC. Patient is a textbook fit if BRCA1 resolves to germline; somatic-only is off-label for the breast indication.

### Wang/Maeda (2016) — Breast Cancer Research

**Reference:**PMID 26772709 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Pten/Trp53 (R270H) GEMM (PTEN-loss basal-like / claudin-low TNBC model)
- **Indication / model:** WAP-Cre;Pten^f/f;p53^lox-stop-lox_R270H composite mice
- **Design:** Mammary-epithelium-specific Pten deletion combined with a TP53-R270H gain-of-function mutation drives aggressive claudin-low and basal-like TNBC with constitutive PI3K/AKT pathway activation
- **n:** Tumour cohort n=20+; transcriptomic profiling n=12
- **Effect size:** moderate — Tumours clustered with the claudin-low / basal-like human TNBC subtypes by transcriptomic profiling, with constitutive AKT activation tracking PTEN loss. The mechanistic preclinical foundation for the PTEN-loss + p53-mutant + AKT-axis subset this patient sits within.
- **Variance:** vs Single-conditional WAP-Cre;Pten^f/f and p53^R270H alone; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Mutation-specific (R270H) p53 allele rather than the broader spectrum of TP53 hotspots in basal-like TNBC; the AKT-inhibitor pharmacology arms come from separate Lin / Davies papers.

### da Motta/Buffa (2016) — Oncogene

**Reference:**PMID 27292261 · **Type:** preclinical · **Inclusion:** excluded — The most directly TNBC-relevant JQ1 paper centres on hypoxia and CA9 angiogenesis rather than the MYC-amplification rationale this case actually carries; the canonical MYC-amplified BET-inhibitor preclinical story is medulloblastoma (PMID 24297863) and the TNBC MYC-amp link is largely in-vitro. The active clinical-trial row (NCT05422794, ZEN-3694 + pembro + nab-paclitaxel) already anchors the BET-inhibitor axis in the case; adding a tangentially-aligned in-vitro CA9 paper would clutter the dossier.

- **Intervention:** JQ1 / BET inhibitors in TNBC (hypoxia / CA9 axis)
- **Indication / model:** HCC1806 (MYC-amplified TNBC) and MDA-MB-231 cell lines and xenografts
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: The most directly TNBC-relevant JQ1 paper centres on hypoxia and CA9 angiogenesis rather than the MYC-amplification rationale this case actually carries; the canonical MYC-amplified BET-inhibitor preclinical story is medulloblastoma (PMID 24297863) and the TNBC MYC-amp link is largely in-vitro. The active clinical-trial row (NCT05422794, ZEN-3694 + pembro + nab-paclitaxel) already anchors the BET-inhibitor axis in the case; adding a tangentially-aligned in-vitro CA9 paper would clutter the dossier. — Worth revisiting if a MYC-amplified TNBC-specific BET-inhibitor xenograft paper surfaces; the current preclinical literature for BET-i in MYC-amp TNBC is weaker than the CDK7 line of evidence (Wang 2015, Patel 2018).

### Sikov WM/Winer EP (2015) — J Clin Oncol

**Reference:**PMID 25092775 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Carboplatin added to neoadjuvant paclitaxel + AC
- **Indication / model:** Stage II-III triple-negative breast cancer, neoadjuvant (neoadj); Clinical stage II-III previously untreated TNBC; BRCA status not stratified at enrollment.
- **Design:** Randomized 2x2 factorial open-label phase 2 (CALGB 40603 / Alliance)
- **n:** 443
- **Effect size:** 60% (carbo) vs 44% (no carbo) % — pCR breast (ypT0/is)
- **Variance:** p=0.0018
- **Toxicities:** neutropenia G>=3 (n=221, 22%); thrombocytopenia G>=3 (n=221, 19%); hypertension G>=3 (n=221, 11%)
- **Case fit:** weak
- **Notes:** Neoadjuvant trial in stage II-III TNBC, not metastatic. Included because it established the pCR gain from adding carboplatin to neoadjuvant chemo in unselected TNBC — supports a carboplatin-containing 1L systemic backbone for this M1 patient given her BRCA1 status.

### Wang/Young (2015) — Cell

**Reference:**PMID 26406987 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** CDK7 inhibitor THZ1 (transcriptional addiction axis)
- **Indication / model:** TNBC cell lines (MDA-MB-231, SUM149, SUM159, HCC70, HCC1806), cell-line and patient-derived TNBC xenografts
- **Design:** Covalent CDK7 inhibition disrupts RNA Pol II CTD phosphorylation at super-enhancer-driven loci, collapsing a TNBC-specific cluster of transcription-addicted genes (the Achilles cluster including MYC, EGFR, FOSL1)
- **n:** n=5 PDX models with n=8 mice/arm; cell-line panel ~14 TNBC lines
- **Effect size:** strong — TNBC cells were hypersensitive to THZ1 with IC50 < 70 nM; in vivo THZ1 induced regression of both cell-line and patient-derived TNBC xenografts while sparing HR+ comparators. The

foundational paper for CDK7 inhibition (samuraciclib, SY-5609) in TNBC including MYC-amplified subsets.

- **Variance:** vs Vehicle; matched HR+/HER2+ lines as comparator; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** THZ1 is a tool compound; clinical translation flowed through samuraciclib (CT7001) and SY-5609. Samuraciclib's Phase 1 TNBC monotherapy signal has been modest, suggesting the in-vivo activity overstates single-agent translation.

### Cardillo/Goldenberg (2015) — Oncotarget

**Reference:**PMID 26101915 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Sacituzumab govitecan (IMMU-132)
- **Indication / model:** TROP2-expressing solid tumor xenografts including MDA-MB-468 (TNBC), pancreatic, gastric, colorectal lines
- **Design:** TROP2-targeted internalisation delivers SN-38 (topoisomerase-I poison) at 136-fold higher tumor concentrations than irinotecan; the moderately stable carbonate linker releases SN-38 both intracellularly and in the tumor microenvironment, producing a bystander effect
- **n:** n=10 mice/arm across MDA-MB-468 and other TROP2+ xenografts
- **Effect size:** strong — MDA-MB-468 (TNBC) xenografts showed complete responses at clinically-relevant SN-38 equivalent doses; the conjugate delivered >100-fold more SN-38 to tumor than free irinotecan. The preclinical foundation for ASCENT and every subsequent TROP2-ADC indication.
- **Variance:** vs Unconjugated hRS7 anti-Trop2 antibody; irinotecan free-drug equivalent doses; non-targeted IgG-SN-38 conjugate; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Companion paper (Cardillo 2011, Clin Cancer Res PMID 21558407) covers the original characterization; the 2015 paper is the multi-tumor extension.

### Fritsch/Maira (2014) — Molecular Cancer Therapeutics

**Reference:**PMID 24652788 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Alpelisib (BYL719, PI3K-alpha inhibitor)
- **Indication / model:** Panel of PIK3CA-mutant and PIK3CA-wild-type cell lines and xenografts; PIK3CA-H1047R knock-in models
- **Design:** Selective ATP-competitive inhibition of the p110-alpha catalytic subunit; PIK3CA hotspot mutants are p110-alpha-addicted and respond preferentially
- **n:** n=8-10 mice/arm in xenograft
- **Effect size:** strong — PIK3CA-mutant breast and gynaecologic lines were 10- to 30-fold more sensitive to BYL719 than wild-type counterparts; xenograft regression in PIK3CA-H1047R tumors at 25 mg/kg. The discovery paper that anchors alpelisib's PIK3CA-hotspot label and the rationale for testing the molecule in PIK3CA-mut TNBC + PTEN-loss settings.
- **Variance:** vs Vehicle; PIK3CA-WT isogenic lines; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** TNBC-specific data thin in this paper; alpelisib's TNBC trial (EPIK-B3 / nab-paclitaxel) halted enrollment in 2022 after slow accrual and an unmet Part B endpoint.

## Shen/Wang (2013) — Clinical Cancer Research

**Reference:**PMID 23881923 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Talazoparib (BMN 673)
- **Indication / model:** Panel of BRCA1/2-mutant cell lines (SUM149, MX-1, CAPAN-1) and isogenic DT40 chicken B-cell HR mutants, plus BRCA2-deficient MX-1 xenografts
- **Design:** Pan-PARP catalytic inhibition plus the most potent PARP-DNA trapping yet described; the trapped lesion is the cytotoxic event in HR-null cells
- **n:** n=6-10 mice/arm in xenograft cohorts; in-vitro SF50 measured in triplicate
- **Effect size:** strong — In SUM149 (BRCA1-mutant TNBC) the SF50 was 8 x 10^-12 mol/L — about 100,000-fold below veliparib. Oral BMN 673 induced complete regression of MX-1 BRCA2-null xenografts at low mg/kg doses, anchoring the EMBRACA dose rationale.
- **Variance:** vs BRCA-wild-type lines; veliparib head-to-head; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong

## Lin/Friedman (2013) — Clinical Cancer Research

**Reference:**PMID 23752188 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Ipatasertib (GDC-0068, pan-AKT inhibitor) and capivasertib (AZD5363)
- **Indication / model:** Panel of >100 cancer cell lines plus xenografts including PTEN-null and PIK3CA-mutant breast, prostate and gastric tumours
- **Design:** ATP-competitive pan-AKT inhibition collapses PI3K/AKT/mTOR signalling; tumors with PTEN loss or PIK3CA hotspot mutations are AKT-dependent and respond preferentially
- **n:** n=8-10 mice/arm in xenograft confirmation
- **Effect size:** moderate — PTEN-null and PIK3CA-mutant lines were enriched among ipatasertib responders, with consistent xenograft regressions in PTEN-null backgrounds. This preclinical signature is what powered IPATunity130 and the PAKT signal in PIK3CA/AKT1/PTEN-altered TNBC.
- **Variance:** vs Vehicle and PTEN/PIK3CA wild-type isogenic lines; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Companion capivasertib preclinical paper (Davies 2012, Mol Cancer Ther) reports the same PI3K/AKT/PTEN-altered sensitivity pattern with the AstraZeneca asset; the two molecules are mechanistically interchangeable in this context.

## Murai/Pommier (2012) — Cancer Research

**Reference:**PMID 23104724 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** PARP-inhibitor class (mechanism: PARP trapping)
- **Indication / model:** DT40 PARP1/PARP2 knockout panel and human DLD-1 isogenic lines treated with olaparib, niraparib, veliparib
- **Design:** Catalytic inhibition is necessary but not sufficient for cytotoxicity; trapping PARP1/PARP2 onto DNA generates the cytotoxic adduct, and trapping potency varies markedly across inhibitors
- **n:** n=3 biological replicates
- **Effect size:** strong — Niraparib > olaparib >> veliparib for PARP-DNA trapping at equimolar exposures, and

trapping correlated with cytotoxicity better than catalytic IC50. This is the paper that explains why talazoparib (highest trapping in the class) outperforms veliparib clinically.

- **Variance:** vs PARP1-null DT40 cells (genetic loss-of-function reference); translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Mechanistic rather than tumor-model paper; the cell systems are not breast lines. Translatability comes from the fact that this trapping hierarchy drives drug-class choice between olaparib and talazoparib.

### Juvekar/Cantley (2012) — Cancer Discovery

**Reference:**PMID 22915751 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** PI3K inhibitor + PARP inhibitor combination (BKM120 / alpelisib + olaparib)
- **Indication / model:** MMTV-CreBrca1(f/f)Trp53(+/-) GEMM of BRCA1-related breast cancer and human BRCA1-mutant TNBC PDX
- **Design:** PI3K inhibition transcriptionally downregulates BRCA1 and ERK1/2-dependent HR factors, exacerbating the HR deficit in BRCA1-related tumors and amplifying olaparib-induced DNA damage
- **n:** GEMM cohort n=10-15 per arm; PDX n=8 per arm
- **Effect size:** strong — Olaparib alone modestly slowed growth in the Brca1-deficient GEMM; the BKM120 + olaparib combination delayed tumour doubling beyond 70 days in the GEMM and beyond 50 days in BRCA1-mutant patient-derived xenografts. This is the canonical preclinical rationale for the alpelisib + olaparib clinical-trial axis (Konstantinopoulos / Mateo).
- **Variance:** vs Vehicle, BKM120 alone, olaparib alone; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** BKM120 (buparlisib) had a CNS-toxicity profile that retired the molecule; alpelisib carries the rationale forward with cleaner pharmacology but with hyperglycemia as the dose-limiting toxicity.

### Bartolomé/Teixidó (2012) — International Journal of Cell Biology

**Reference:**PMID 22245967 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** CXCR4 / CXCL12 axis in breast cancer liver tropism
- **Indication / model:** MDA-MB-231 and MDA-MB-468 TNBC cells in a rat intravital-microscopy liver metastasis model; CXCR4 shRNA knockdown
- **Design:** Hepatic sinusoidal endothelium and stellate cells secrete CXCL12 (SDF-1), creating a chemotactic gradient that arrests CXCR4-positive breast cancer cells in the liver microvasculature and drives transendothelial extravasation
- **n:** n=6-8 rats/arm in intravital experiments; multiple biological replicates of shRNA knockdown
- **Effect size:** moderate — CXCR4 knockdown reduced hepatic arrest and extravasation of MDA-MB-231 cells in real-time intravital imaging; pharmacological CXCR4 blockade with AMD3100 phenocopied the genetic effect. Mechanistic foundation for why TNBC homes to the liver and a hypothesis-generating row for the patient's solitary-hepatic-lesion presentation.
- **Variance:** vs Scrambled shRNA; CXCR4-intact parental cells; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Rat model and basic-biology paper rather than therapeutics; no CXCR4 antagonist is on the table for this patient. Included as mechanism context for the oligomet-liver pattern, not as a drug rationale.

### Lehmann/Pietenpol (2011) — Journal of Clinical Investigation

**Reference:**PMID 21633166 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Lehmann TNBCtype molecular subtyping (refines basal-like into BL1/BL2/M/MSL/LAR/IM)
- **Indication / model:** Expression-profile meta-analysis of 587 TNBC samples across 21 datasets; preclinical drug sensitivity in matched cell lines
- **Design:** TNBC partitions into transcriptomically distinct subtypes with divergent therapeutic vulnerabilities — BL1 (cell-cycle / DDR), BL2 (growth factor / glycolysis), M / MSL (EMT, mesenchymal), LAR (androgen-receptor), IM (immune)
- **n:** 587 TNBC tumours; cell-line panel n=25
- **Effect size:** moderate — BL1 (DDR-enriched) cell lines were the most cisplatin-sensitive; LAR lines responded to bicalutamide; M/MSL lines to dasatinib. The platform paper that lets the team interpret the patient's basal-like label at therapeutic resolution and underpins target_validation row pam50-subtype-confirmation.
- **Variance:** vs Not applicable — classifier development with in-vitro drug-sensitivity validation; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Classifier-and-cell-line-panel paper; no in-vivo data. Drives the rationale for the PAM50 + Lehmann TNBCtype-4 confirmatory test already queued in target_validation.

### Hirai/Iwasawa (2009) — Molecular Cancer Therapeutics

**Reference:**PMID 19887545 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Adavosertib (AZD1775 / MK-1775, WEE1 inhibitor)
- **Indication / model:** p53-deficient human tumor lines (lung, breast, colon) and matched xenografts; combination with gemcitabine, carboplatin, cisplatin, 5-FU
- **Design:** WEE1 inhibition removes the G2/M checkpoint that p53-null tumors depend on after DNA damage, forcing premature mitotic entry and mitotic catastrophe
- **n:** n=6-8 mice/arm xenograft
- **Effect size:** strong — MK-1775 selectively sensitized p53-deficient tumour cells to gemcitabine, carboplatin and cisplatin both in vitro and in xenografts, while sparing p53-wild-type cells. The foundational paper for the TP53-mutant WEE1-inhibitor hypothesis that applies to ~85% of basal-like TNBC including this patient.
- **Variance:** vs p53-wild-type isogenic clones; chemo-only arms; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Breast lines were a minor part of the panel; the TNBC-specific extension comes from later work (Pitts 2020, capecitabine combo). Clinical adavosertib development was discontinued in 2021 over tolerability; debio-0123 and next-generation WEE1 inhibitors carry the rationale forward.

### Saal/Hennessy (2009) — Cancer Research

**Reference:**PMID 19208841 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** PI3K/AKT axis biology in basal-like TNBC (PTEN-loss landscape)
- **Indication / model:** Panel of 33 breast cancer cell lines (incl. basal-like TNBC HCC1937, MDA-MB-468, HCC70, BT-549) profiled for PTEN status, RPPA pathway readout, and drug sensitivity
- **Design:** Basal-like TNBC harbours frequent PTEN genomic loss and concomitant PI3K/AKT pathway

hyperactivation, often co-occurring with BRCA1 deficiency in the same tumours

- **n:** n=33 cell lines profiled
- **Effect size:** moderate — Basal-like TNBC cells expressed significantly lower PTEN than luminal counterparts and showed correspondingly higher AKT pathway activation; PTEN loss tracked with sensitivity to PI3K/AKT-axis inhibition. Sets up the biomarker case for capivasertib + paclitaxel in PIK3CA/AKT1/PTEN-altered TNBC (PAKT, IPATunity130, CAPItello-290).
- **Variance:** vs Luminal / HER2+ comparator lines within the panel; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Cell-line panel paper; no in-vivo pharmacology. Directly supports the patient's PTEN-loss + PIK3CA-mut + basal-like archetype, but the therapy connection runs through later xenograft papers.

### Sakai/Taniguchi (2008) — Nature

**Reference:**PMID 18264088 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** PARP inhibitors / platinum (resistance via BRCA reversion)
- **Indication / model:** BRCA2-mutant CAPAN-1 pancreatic line treated with cisplatin to select resistance; sequencing of revertants
- **Design:** Secondary intragenic frameshift-restoring mutations in BRCA2 re-open the reading frame and restore functional protein, rescuing HR and conferring resistance to platinum and (later shown) PARP inhibitors
- **n:** 13 cisplatin-resistant clones characterised
- **Effect size:** strong — Cisplatin-resistant CAPAN-1 clones acquired secondary BRCA2 mutations that restored the reading frame and functional protein. The mechanistic ancestor of every ctDNA-reversion paper in BRCA-mutant breast and ovarian cancer.
- **Variance:** vs Parental cisplatin-sensitive CAPAN-1; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** BRCA2 not BRCA1; the BRCA1 reversion analog (Edwards/Norquist) is conceptually identical and reported in the same year. Justifies the baseline ctDNA row in target_validation.

### Liu/Berns (2007) — PNAS

**Reference:**PMID 17626182 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Brca1/Trp53 conditional knockout GEMM (basal-like TNBC model platform)
- **Indication / model:** K14cre;Brca1^F/F;Trp53^F/F conditional double-knockout mouse mammary tumors
- **Design:** Tissue-specific somatic loss of Brca1 plus Trp53 in basal mammary epithelium recapitulates the genomic instability, ER-negative status, basal-marker expression, and copy-number landscape of human BRCA1-mutated basal-like breast cancer
- **n:** Cohort of >20 mice followed to tumor endpoint; tumor characterisation on n>15 tumors
- **Effect size:** strong — Tumours were poorly differentiated, ER-negative, basal-marker positive, and genomically unstable — the closest GEMM faithful replication of the patient's BRCA1-mut TP53-mut basal-like TNBC phenotype. This is the workhorse platform for PARP-inhibitor, PARP+PI3K (Juvekar), and immune-combination preclinical studies in BRCA1-related TNBC.
- **Variance:** vs K14cre;Trp53^F/F single-conditional and K14cre alone; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong

- **Notes:** Model platform paper rather than a therapy paper. Adding PTEN loss requires the Wartman 2012 or Liu 2007 follow-up that intercrosses Pten conditional alleles.

### Bryant/Helleday (2005) — Nature

**Reference:**PMID 15829967 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Olaparib (and the PARP-inhibitor class)
- **Indication / model:** BRCA2-deficient V-C8 hamster cells, BRCA2-mutant CAPAN-1 pancreatic line, and isogenic BRCA1-knockdown MCF-7 breast cancer cells
- **Design:** PARP1 catalytic inhibition leaves single-strand breaks unrepaired; these collapse to double-strand breaks at the replication fork, which BRCA-deficient cells cannot resolve through homologous recombination
- **n:** n=3 biological replicates per cytotoxicity curve; xenograft arm n=8 mice
- **Effect size:** strong — BRCA1- and BRCA2-deficient cells were 1,000-fold more sensitive to PARP inhibition than isogenic wild-type counterparts, with KU0058684 inducing xenograft regression in BRCA2-null tumours. One of the two foundational synthetic-lethality papers underpinning every BRCA + PARPi indication.
- **Variance:** vs BRCA-wild-type sister cell lines and vehicle-treated xenografts; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** BRCA2-led data with BRCA1 confirmation in MCF-7 knockdown rather than a BRCA1-mutant TNBC line; subsequent papers extended the phenotype to HCC1937 and MDA-MB-436.

### Farmer/Ashworth (2005) — Nature

**Reference:**PMID 15829966 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Olaparib (and the PARP-inhibitor class)
- **Indication / model:** BRCA1-deficient and BRCA2-deficient embryonic stem cells, HCC1937 (BRCA1 5382insC) and isogenic reconstituted lines
- **Design:** Loss of BRCA1/2 abolishes HR repair of replication-associated double-strand breaks; persistent PARP-bound DNA lesions accumulate when PARP catalysis is blocked, killing HR-null cells selectively
- **n:** n=3 biological replicates per dose-response
- **Effect size:** strong — HCC1937 BRCA1-null cells were ~1,000-fold more sensitive to KU0059436 than reconstituted controls. This is the second of the paired 2005 Nature papers and the one that explicitly used HCC1937 — the BRCA1-mutant TNBC line most relevant to this patient.
- **Variance:** vs BRCA1/2-reconstituted isogenic clones; wild-type ES cells; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong